

## COP3330C Module 9 Programming Assignment 1: Applying the Adapter Pattern with Java Collections

In this assignment, you will solidify your understanding of Java collections, generics, and design patterns by creating a Java program that demonstrates the use of the Adapter pattern, along with other Java concepts such as the Diamond Operator, sorting a list of numbers, using the remove() method, and employing unbounded wildcards. Note that this assignment requires the submission of a source code file and a screenshot of command line execution (not from an IDE).

### Instructions:

#### 1. Setup Your Project:

- Create a new Java class named ConceptDemo.
- Write a main() method to implement all of the required tasks.

#### 2. Apply the Adapter Pattern:

- Implement an Adapter to adapt an array of Integer values into a List<Integer> interface.
- Create an Integer[] array with several elements (e.g., 5, 2, 9, 1, 4, 7).
- Use the adapter to convert the array into a List<Integer>.

Note that we are using an **Object Adapter** here (as opposed to a Class Adapter); the Integer array is used as a component to create a List that adapts the array

#### 3. Use the Diamond Operator:

- Create an ArrayList of Integer using the Diamond Operator (<>) for type inference.
- Add additional elements to the list using Collections.addAll().

#### 4. Sort the List:

- Print the list before sorting.
- Use the Collections.sort() method to sort the elements of the list.
- Print the sorted list to verify the results.

#### 5. Use the remove() Method:

- Remove a specific element from the list (e.g., remove the value 4).
- Print the list again to show the result after the removal.

#### 6. Demonstrate Unbounded Wildcards:

- Create a method named printList() that takes a list of any type (List<?>) as a parameter.
- Use this method to print the list's elements.

#### 7. Testing:

- Your program should print the following:
  - The list before sorting.
  - The list after sorting.
  - The list after removing an element.
  - The list printed using the printList() method with an unbounded wildcard.

**Program Output Example:**

```
Before sorting: [5, 2, 9, 1, 4, 7]
After sorting: [1, 2, 4, 5, 7, 9]
After removing 4: [1, 2, 5, 7, 9]
Printing list using unbounded wildcard: [1, 2, 5, 7, 9]
```

**Functional Requirements:**

- Adapter Pattern: Correctly adapt an Integer[] array to a List<Integer>.
- Diamond Operator must be used for list creation.
- Sorting must be demonstrated using the Collections.sort() method.
- Remove an element from the list by specifying the value.
- Unbounded wildcard must be used in the printList() method.

**Non-Functional Requirements:**

- 4-line ID Header (top of the source file(s)):
  - // FileName.java
  - // your name
  - // submission date
  - // one-line summary of program's purpose
- All lines <= 80 columns (including comments)
- Useful comments
- Correct indentation (minimum of 3 spaces per level)

**Total Points: 20**

**Submission:**

- Submit your ConceptDemo.java file to the GitHub repository. Do not submit .class files.
- Submit a screenshot of the program output demonstrating the steps as shown above. The screenshot must be taken of execution from the command line, not from your IDE. It should be clear and legible.

**Hints:**

- Be careful to use Integer.valueOf() when removing elements from the list to avoid ambiguity between removing by index and removing by value.
- Use appropriate comments in your code to explain each major step.
- For the Adapter pattern, you can create a utility function that takes an Integer[] array parameter and returns a List<Integer>.

**Grading:**

- Correct implementation of Adapter Pattern (5 points)
- Correct Usage of Diamond Operator (2 points)
- Correct Sorting Implementation (2 points)

- Proper Use of remove() Method (2 points)
- Implementation of Unbounded Wildcard Method (2 points)
- Code Readability and Comments (2 points)
- Clear and legible screenshot (5 points)